

Problem 1 : l^∞ as a Complete Metric Space

Let $l^\infty = l^\infty(\mathbb{N})$ denote the set of all bounded complex-valued sequences $x = (x_n)_{n=1}^\infty$.

(a) Show that l^∞ is a complete metric space when equipped with the metric

$$d(x, y) = \sup_n |x_n - y_n|.$$

(b) Let $(\varepsilon_n)_{n=1}^\infty$ be a sequence of positive real numbers such that $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. Show that

$$M = \{x = (x_n) \in l^\infty : |x_n| \leq \varepsilon_n, \quad n = 1, 2, \dots\}$$

is a compact subset of l^∞ .

Solution. (a): Let $(x^m)_{m \in \mathbb{N}} \subset l^\infty$ be a sequence of bounded complex valued sequences that is Cauchy. Thus $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $m_1 \geq m_2 \geq N$: $d(x^{m_1}, x^{m_2}) < \varepsilon$. This means $\forall n \in \mathbb{N}$,

$$|x_n^{m_1} - x_n^{m_2}| \leq d(x^{m_1}, x^{m_2}) < \varepsilon.$$

Thus for fixed n , $(x_n^m)_{m \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{C}$. By Completeness of $\mathbb{C}$,

$$\lim_{m \rightarrow \infty} x_n^m$$

exists. Let

$$y_n := \lim_{m \rightarrow \infty} x_n^m.$$

We want to show two things regarding y :

1. y bounded: $\exists N \geq 0$ such that $d(x^m, x^n) < 1$ for $m, n \geq N$ this means $\forall n \in \mathbb{N}$: $|x_n^p - x_n^N| < d(x^m, x^N) < 1$. Thus

$$|y_n| \leq |y_n - x_n^N| + |x_n^N| = \lim_{p \rightarrow \infty} |x_n^p - x_n^N| + |x_n^N| \leq 1 + d(x^N, 0)$$

Thus $d(y, 0) < \infty$.

2. $x^m \rightarrow y$ in l^∞ : $\forall \varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\forall m \geq n \geq N$: $d(x^m, x^n) < \frac{\varepsilon}{2}$. $\forall p \in \mathbb{N}$:

$$|x_p^m - x_p^n| < d(x^m, x^n) < \frac{\varepsilon}{2} \Rightarrow |y_p - x_p^n| = \lim_{m \rightarrow \infty} |x_p^m - x_p^n| \leq \frac{\varepsilon}{2}.$$

Thus $d(y, x^n) \leq \frac{\varepsilon}{2} < \varepsilon$. Thus $x^m \rightarrow y$ as $m \rightarrow \infty$.

(b): Let $(x^m)_{m \in \mathbb{N}} \subset M$ we want to show that x^m has a convergent subsequence. We know for all $m \in \mathbb{N}$ and for all $n \in \mathbb{N}$, $|x_n^m| \leq \varepsilon_n$. Thus $\forall n \in \mathbb{N}$ the sequence $(x_n^m)_{m \in \mathbb{N}}$ is a bounded sequence. So for fixed $n \in \mathbb{N}$ there exists a subsequence $(x_n^{m_i})_{i \in \mathbb{N}}$ that is convergent. So here is what you do, starting with $n = 1$ you filter out such that $(x_n^m)_{m \in \mathbb{N}}$ is a convergent sequence and then you increment n and filter out so that for all n $(x_n^m)_{m \in \mathbb{N}}$ is a convergent sequence.

Thus wlog the sequence x^m converges point wise to a sequence $y \in l^\infty$. Now we have to show $x^m \rightarrow y$ as $m \rightarrow \infty$ in l^∞ .

$\forall \varepsilon > 0$, there exists a $K \in \mathbb{N}$ such that $\varepsilon_n < \frac{\varepsilon}{2}$ for all $n \geq K$. Thus for all m :

$$|y_n - x_n^m| \leq |y_n| + |x_n^m| \leq \varepsilon_n + \varepsilon_n < \varepsilon.$$

Now we have to show that for the first $K - 1$ we can pick a M large enough that $\forall m \geq M$: $|y_k - x_k^m| < \varepsilon$ for $k \in [K - 1]$. We can do this because $x_k^m \rightarrow y_k$ for all k as $m \rightarrow \infty$. For each $k \in [K - 1]$, there exists $M_k \in \mathbb{N}$ such that $\forall m \geq M_k$: $|y_k - x_k^m| < \varepsilon$. Let $M = \max\{M_k : k \in [K - 1]\}$. Thus $\forall m \geq K$: $|y_k - x_k^m| < \varepsilon$.

Thus $\forall m \geq M$ and all $n \in \mathbb{N}$:

$$|y_n - x_n^m| < \varepsilon \Rightarrow d(y, x^m) \leq \varepsilon.$$

Thus $x^m \rightarrow y$ as $m \rightarrow \infty$ in l^∞ . Thus our original sequence has a convergent subsequence. Hence M is compact.

Alternative Proof: In part (a) we proved that l^∞ was complete. So to prove that M is compact it suffices to show that its closed and totally bounded.

1. Totally Bounded: $\forall \varepsilon > 0$, by convergence of $\varepsilon_n \rightarrow 0$ there exists a $N \in \mathbb{N}$ such that $\forall n \geq N$: $\varepsilon_n < \frac{\varepsilon}{4}$. Thus $\forall x, y \in M$

$$|x_n - y_n| \leq 2\varepsilon_n < \frac{\varepsilon}{2} < \varepsilon.$$

. We now need to show that there exists a collection $\alpha^1, \dots, \alpha^k \in M$ such that $\forall x \in M$ there exists a $m \in \{1, \dots, k\}$ such that for all $n \in \{1, \dots, N - 1\}$ we have that

$$|x_n - \alpha_n^m| < \frac{\varepsilon}{2} \Rightarrow d(x, \alpha^m) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Note the following: $\forall x \in M$, $x_{1, \dots, N-1} = (x_1, \dots, x_{N-1}) \in \{x \in \mathbb{C}^{N-1} : |x_n| \leq \varepsilon_n\} = A$. Note A is closed and bounded and hence compact in $\mathbb{C}^{N-1}$. By total boundedness of compact sets, there exists $y^1, \dots, y^k \in A$ such that

$$A = \bigcup_{i=1}^k B_{\frac{\varepsilon}{2}}(y^i)$$

Define α^i as follows:

$$\alpha_j^i := \begin{cases} y_j^i & \text{if } j < N \\ 0 & \text{otherwise} \end{cases}.$$

By definition $\alpha^i \in M$. Furthermore for all $x \in M$, note there exists a $i \in \{1, \dots, k\}$ such that for $n \in \{1, \dots, N - 1\}$:

$$|x_n - \alpha_n^i| = |x_n - y_n^i| \leq d_{\mathbb{C}^{N-1}}(x_{1, \dots, N-1}, y^i) < \frac{\varepsilon}{2} < \varepsilon$$

.

Thus $\forall n \in \mathbb{N}$:

$$|x_n - \alpha_n^i| < \frac{\varepsilon}{2} \Rightarrow d(x, \alpha^i) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Thus

$$M = \bigcup_{i=1}^k B_\varepsilon(\alpha^i)$$

2. Closed: Let (y^i) be a convergence sequence in M where $y^i \rightarrow y$. Since for all i and all n : $|y_n^i| \leq \varepsilon_n$, y respects the same inequality and hence is in M .

Problem 2 : Isometry on a Compact Metric Space

Let (X, d) be a compact metric space and let $f : X \rightarrow X$ be an isometry (meaning that $d(f(x), f(y)) = \rho(x, y)$ for all $x, y \in X$). Prove that f is a bijection.

Hint. Consider $x \notin f(X)$ and follow the iterates of f on x .

Solution. (a): Surjectivity: Assume the contrary $f(X) \neq X$. Thus there exists $x \in X \setminus f(X)$. Create the sequence $x_0 = x$ and $x_n = f(x_{n-1})$. For $m > n$:

$$\begin{aligned} d(x_m, x_n) &= d(f^n(x_{m-n}), f^n(x_0)) \\ &= d(x_{m-n}, x_0). \end{aligned}$$

Take $\varepsilon = d(x_0, f(X))$. Note that $\forall i \in \mathbb{N}$, $d(x_i, x_0) \geq \varepsilon$.

By the sequential definition of compactness, there exists a subsequence x_{m_i} which is convergent hence Cauchy. Thus there exists $N \in \mathbb{N}$ such that for all $j \geq i \geq N$:

$$d(x_{m_j - m_i}, x_0) = d(x_{m_j}, x_{m_i}) < \varepsilon.$$

Thus

$$\varepsilon \leq d(x_{m_i - m_N}, x_0) = d(x_{m_j}, x_{m_i}) < \varepsilon.$$

Thus a contradiction occurs.

Alternative Proof: By Total Boundedness of X , $\forall \varepsilon > 0$ there exists $A = \{a_1, \dots, a_k\} \subset X$ where

$$X = \bigcup_{i=1}^k B_\varepsilon(a_i).$$

Without loss of generality assume $d(a_i, a_j) > \varepsilon$ and that k is the maximum number of such points.

(b): Injectivity: Suppose $f(x) = f(y)$. Then $d(x, y) = d(f(x), f(y)) = 0 \Rightarrow x = y$.

Problem 3 : Diameter of Nested Closed Sets

Let X be a compact metric space and suppose that for every $n = 1, 2, \dots$, F_n is a closed non-empty subset of X such that $F_n \supseteq F_{n+1}$. Show that

$$d\left(\bigcap_{n=1}^{\infty} F_n\right) = \inf_n d(F_n).$$

Here

$$d(A) = \sup_{x, y \in A} d(x, y)$$

is the diameter of $A \subseteq X$.

Solution. Since for all n

$$\bigcap_{n \in \mathbb{N}} F_n \subset F_n \Rightarrow d\left(\bigcap_{n \in \mathbb{N}} F_n\right) \leq d(F_n),$$

this means

$$d\left(\bigcap_{n \in \mathbb{N}} F_n\right) \leq \inf_n d(F_n).$$

We want to show that

$$\inf_n d(F_n) \leq d\left(\bigcap_{n \in \mathbb{N}} F_n\right).$$

Create sequences x_n and y_n where $x_i, y_i \in F_i$ such that $d(x_i, y_i) = d(F_i)$ (this is possible because F_i is compact and thus achieves its maximum. Note F_1 is closed hence compact, thus by sequential definition of compactness, there exists a subsequence which is convergent. Pass to the subsequences of x_n and y_n , thus $x_n \rightarrow x$ and $y_n \rightarrow y$. Note the following $\forall n \in \mathbb{N}$, $x_{i+n}, y_{i+n} \in F_n$ for all $i \in \mathbb{N}$ and thus by the closedness of F_n , $x, y \in F_n$. Thus $x, y \in F_n$ for all n and $x, y \in \bigcap F_n$. $d(\cdot, \cdot)$ is continuous so since $d(x_n, y_n) \rightarrow d(x, y)$ as $n \rightarrow \infty$. Thus

$$\inf_n d(F_n) = d(x, y) \leq d\left(\bigcap_{n \in \mathbb{N}} F_n\right).$$

Thus we get equality as described in the problem.

Problem 4 : Continuity and Compact Subsets

Let X and Y be metric spaces. Show that a map $f : X \rightarrow Y$ is continuous if and only if its restriction to each compact subset of X is continuous.

Solution.

Problem 5 : Subadditive Sequences

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive real numbers such that

$$a_{n+m} \leq a_n + a_m, \quad m, n \geq 1.$$

Show that $\lim_{n \rightarrow \infty} \frac{a_n}{n}$ exists by showing

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}.$$

Hint. Treat separately $\liminf$ and $\limsup$.

Solution.